DECK-KEYS Bukifon v1.0 290 x 78 x 1.6 mm (spec/10-panel-ui.md 3.2, spec/20-electronics.md 6.4) CERN-OHL-S-2.0

42 x Kailh Choc (hot-swap socket) + 42 x 1N4148, matrix 6 rows x 7 columns on MCP23017 (0x20) + 42 x SK6812MINI-E + 2 x 30 mm fader + IDC-16

Matrix 6 rows x 7 cols: ROW1..3 = left half, ROW4..6 = right half, COL1..6 = main columns, COL7 = thumb row. ROWn = MCP23017 GPA0..GPA5 (driven low). COLn = GPB0..GPB6 (read with pull-up). Full KEYn <-> (row,col) <-> panel coordinate table: matrix-tables.md

SL1/SL2 in the central zone (x 110..205, same coordinates as DECK-PANEL), 100 mm slide pots across 3V3. 1k/100n RC on PANEL-1.

DECK_ID for DECK-KEYS: 3.3k / 33k -> 3.3*33/36.3 = 3.00 V (spec/20-electronics.md 6.2: 3.00 V +-0.15 V)

42 x SK6812MINI-E, chain after LED_IN from PANEL-1, powered from +5V (J-DECK pin 16), 100n per LED; last DOUT = no connect.

Matrix 42 keys uses GPA0..5 + GPB0..6; GPA7/GPB7 drive the 2 central-zone mode indicators (1k series, active low).

J-DECK pin 15 (TSR) is not used on DECK-KEYS -> GND: pin 16 = +5V (LED chain).

Bukifon open hardware - CERN-OHL-S-2.0

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Title: Bukifon DECK-KEYS — 290x78 — 42x Kailh Choc hotswap / 42x1N4148 / 42x SK6812MINI-E

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